TABLE OF CONTENTS

1 - OVERVIEW

This section applies to Signa 1.5T systems installed with an SRF Cabinet and explains the procedures for the replacement of the RF amplifiers and the fans in the RF amplifiers.

Parts mentioned are listed in Table 1-1:

TABLE 1-1
FRU(1) PART LIST

Manufacturer's Part Number	GEMS Part Number	Item Description
AN8102G	2224562-5	RF Amplifier with Case
21-515283	2262478	Fan

Special tools needed to replace the amplifier are listed in Table 1-2:

TABLE 1-2
SPECIAL TOOLS NEEDED

Item	Description	Part Number
1	Universal Lift Hoist Kit – Used to remove amp from cabinet.	46-317266G5

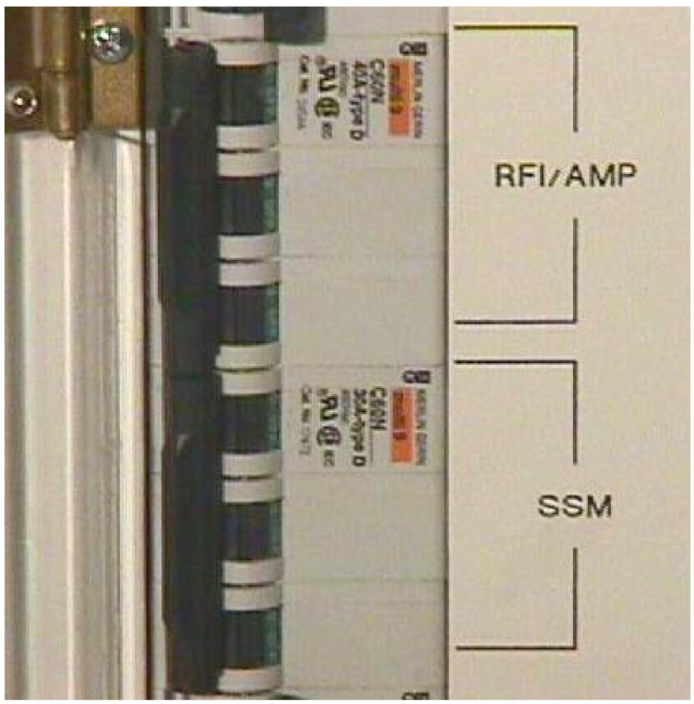
2 – SRF CABINET POWER LOCKOUT AND TAGOUT

DANGER!!

POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRF CABINET BEFORE ATTEMPTING TO REMOVE THE RF AMPLIFIER. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

1. Remove the cover from the front of the ACGD cabinet.

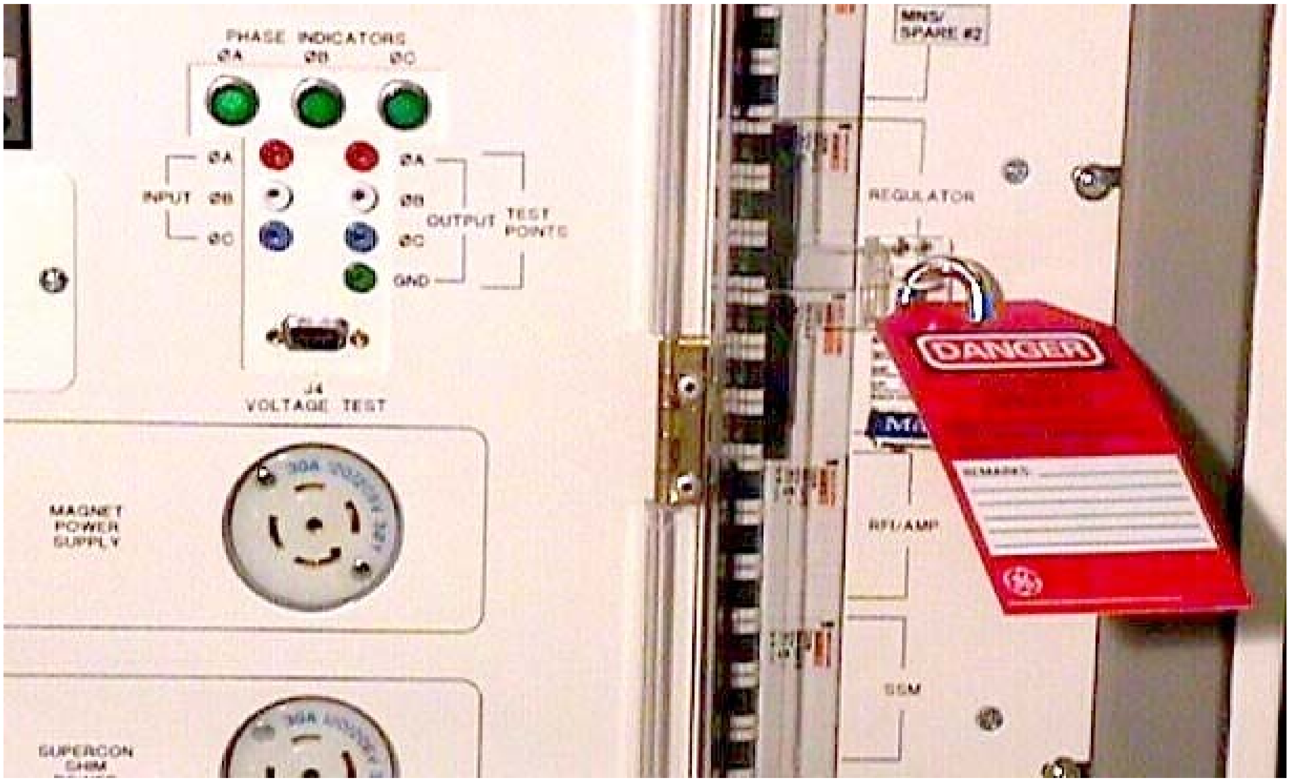
2. Open Plexiglas door over the front of the PDU breakers and switch off the **RFI/AMP** and **SSM** breakers. See Illustration 2-1.



BREAKER IDENTIFICATION
ILLUSTRATION 2-1

2 – SRF CABINET POWER LOCKOUT AND TAGOUT (CONTINUED)

3. Close the plexi-glass door and lock and tag it out. See Illustration 2-2. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools,* Safety*,* Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS*.*)



LOCKOUT AND TAGOUT OF BREAKERS
ILLUSTRATION 2-2

4. Remove the front cover from the SRF Cabinet.

5. Verify all LEDs are OFF (not illuminated) on front of SSM and RFI. Verify all LEDs are OFF (not illuminated) at the rear of the RF Amplifier Power Module.

6. Using an AC voltmeter (DVM), measure the input power leads on the rear of one of the RF Amplifier Modules from the rear (right side) of the SRF Cabinet.

   - Measure between L1 and L2; verify voltmeter measures 0 VAC.

3 - RF AMPLIFIER

3-1 RF Amplifier Power Module Removal

DANGER!!

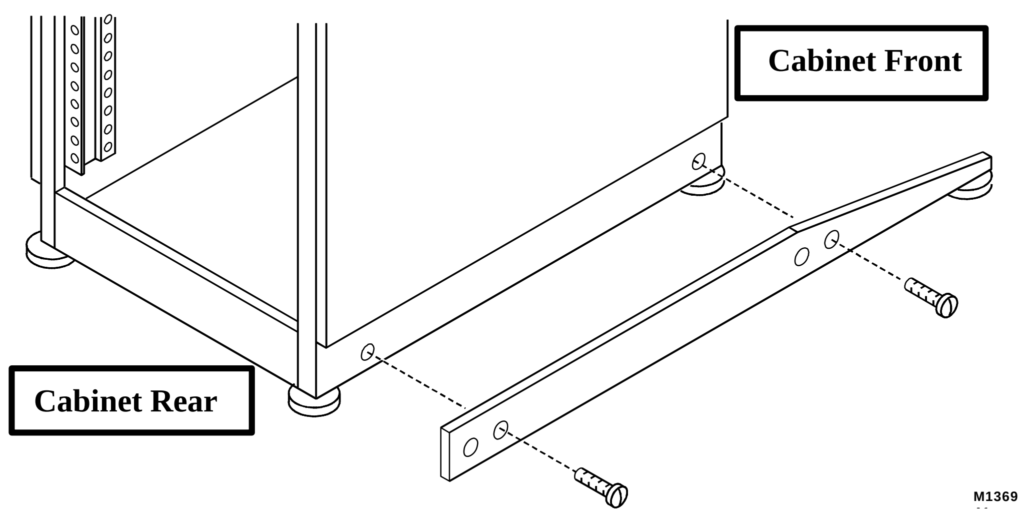
POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRF CABINET BEFORE ATTEMPTING TO REMOVE THE RF AMPLIFIER. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

1. Perform Section **2 – SRF CABINET POWER LOCKOUT AND TAGOUT** (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)

DANGER!!

CABINET TIP HAZARD! POSSIBLE PERSONAL INJURY! CONFIRM THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON EITHER SIDE OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO REMOVE AN RF AMPLIFIER.

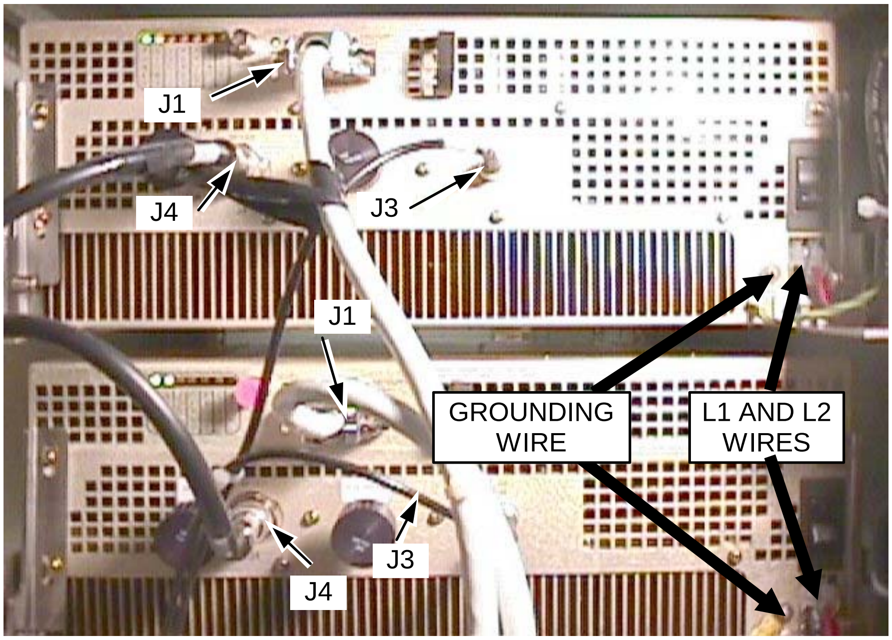
2. Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the SRF Cabinet. See Illustration 3-1.



INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET
ILLUSTRATION 3-1

3-1 RF Amplifier Power Module Removal (Continued)

3. Disconnect wires from L1 and L2 of AC connectors and the grounding wire on rear of RF AMP. See Illustration 3-2.

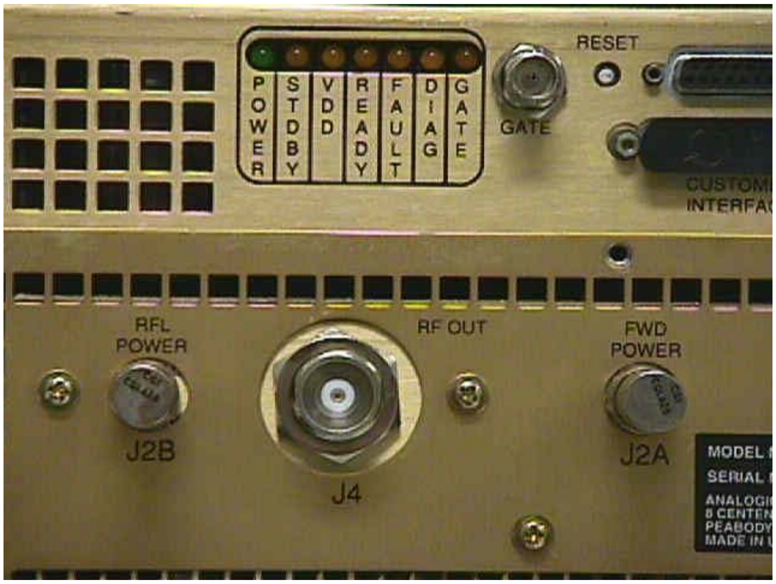


CONNECTOR LOCATIONS
ILLUSTRATION 3-2

4. Remove the J1, J3, and J4 connections on rear of RF Amplifiers. See Illustration 3-2.

3-1 RF Amplifier Power Module Removal (Continued)

5. Remove the two 50 ohm terminators (or dummy loads as seen in Illustration 3-2) from BNC connectors J2A (labeled FWD POWER) and J2B (labeled RFL POWER) on the rear of the amplifier. See Illustration 3-3.



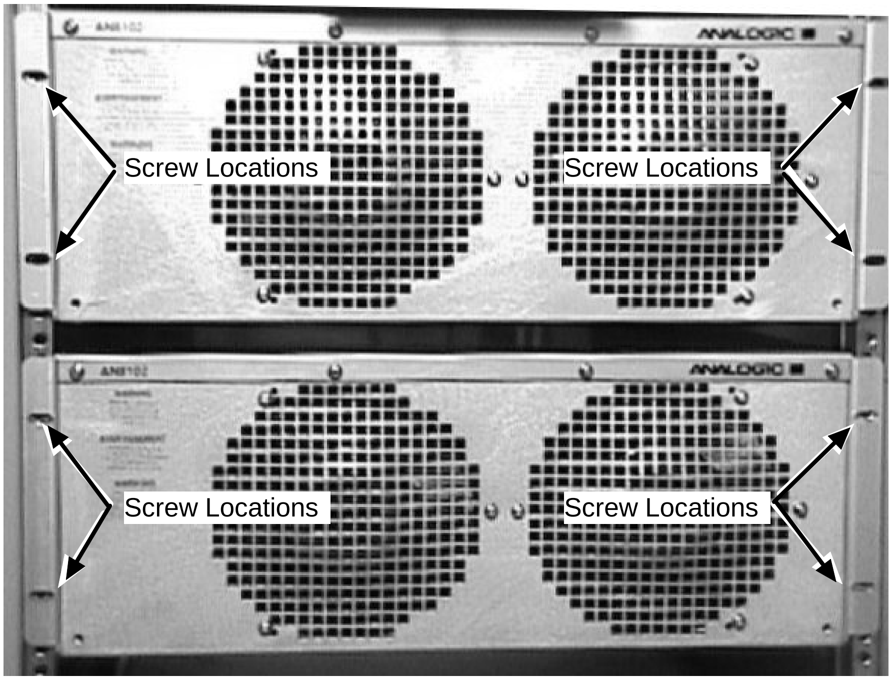
J2B AND J2A CONNECTORS WITH TERMINATORS ON REAR OF AMPLIFIER
ILLUSTRATION 3-3

6. Place these terminators (or dummy loads) into an empty cut-out in the padding inside the FRU container in which the amplifier will be returned. This will prevent damage to the terminators and to the J2A and J2B BNC connectors during shipment. Do not return the amplifier with the terminators (or dummy loads) connected as shown in Illustrations 3-2 and 3-3.

3-1 RF Amplifier Power Module Removal (Continued)

7. Remove the four screws securing the amplifier to the cabinet. See Illustration 3-4.

FRONT VIEW



SCREW LOCATIONS ON CABINET
ILLUSTRATION 3-4

8. Refer to **APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR 1.5T RF/PDU AND SRF CABINETS** procedure to remove amplifier from cabinet.

3-2 Amplifier Installation

1. Refer to **APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR 1.5T RF/PDU AND SRF CABINETS** procedure and install the amplifier into the cabinet.

2. Insert four screws into front of amplifier to secure to cabinet.

3. Reconnect cables J1, J3, and J4 connections on rear of RF Amplifiers

4. Reconnect the 50 ohm terminators (included in a plastic bag shipped with the amplifier) to BNC connectors J2A and J2B on the rear of the amplifier. See Illustration 3-3.

5. Connect the L1 and L2 AC power wires to the AC power connectors on the rear of the amplifier.

6. Connect the grounding wire to the lug on the rear of the RF amplifier.

7. Confirm that the black power rocker switch above the AC power connectors on the rear of the amplifier is in the UP/ON position.

3-2 Amplifier Installation (Continued)

8. Close the SRF rear cabinet door.

9. Remove lock and tag from the plexi-glass cover on the front of the PDU cabinet and put the RFI/AMP and the SSM breakers in the ON position.

10. Confirm that LEDs are illuminated on the front of the RFI and SSM and confirm that the fans in the RF amplifiers are rotating.

11. If an amplifier was replaced then perform the RFI calibration procedure described in **1.5T SRF Max Power RF Out Setup and Calibration** (RC5SCA1.DOC) on the 8.X Service Methods CDROM.

4 - FAN

4-1 Tools Required

- Phillips Screwdriver (30cm or longer)

4-2 Removal of Fans

DANGER!!

SHOCK HAZARD! AVOID SERIOUS INJURY BY ELECTROCUTION. THE CAPACITOR BANK IN THE AMPLIFIER MAY RETAIN CHARGE LONG AFTER THE INCOMING 208V~ L1 AND L2 LINES THAT CONNECT AT THE REAR OF THE AMPLIFIER HAVE BEEN DISCONNECTED. BEFORE REPLACING ANY COMPONENTS IN THE AMPLIFIER, MEASURE THE CAPACITOR VOLTAGE AS DESCRIBED IN THIS SECTION TO VERIFY THAT THE VOLTAGE IS BELOW 10 VDC. DO NOT ASSUME THAT THE BLEEDER RESISTOR HAS DISCHARGED THE CAPACITOR BANK BASED ON THE STATE OF THE INDICATOR LEDS. ALWAYS MEASURE TO VERIFY.

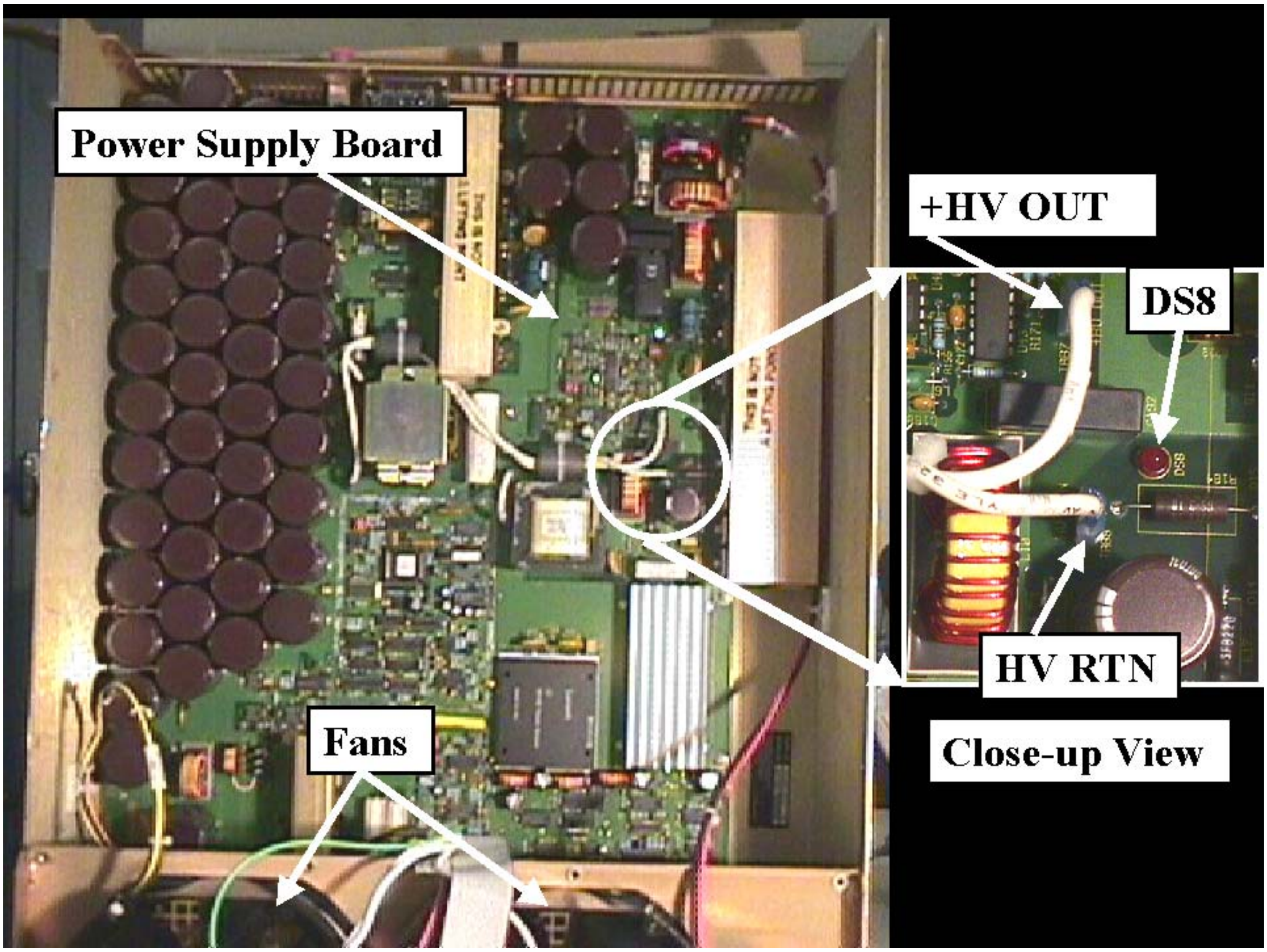
1. Remove RF Amplifier from cabinet. Refer to Section **3-1 RF Amplifier Power Module Removal**.

2. With unit resting on a flat surface, remove 16 screws from the RF AMP.
   - Four screws along front top of RF Amplifier
   - Four screws along rear top of RF Amplifier
   - Four screws along each side of the RF Amplifier

3. Lift RF Amplifier top cover assembly by the handle on the rear of the amplifier.

4-2 Removal of Fans (Continued)

4. Refer to Illustration 4-1. Locate the red LED DS8 that is on the power supply board mounted underneath the top cover. DS8 provides a visual indication of whether there is voltage on the +HV OUT line or not. During normal operation there is 141 VDC present between the +HV OUT and HV RTN lines.



POWER SUPPLY BOARD AND CLOSE-UP OF LED DS8 AND +HV AND HV RTN LINES
ILLUSTRATION 4-1

5. Determine whether LED DS8 is illuminated or not.

   a. If DS8 is illuminated then wait the time necessary for it to eventually go dark. This usually takes about 8 minutes.

      i. After DS8 has gone dark then use a multimeter to measure the voltage between the exposed portions of the spade connectors of the +HV and HV RTN lines. Do not remove the wires from the spade connectors. Leave the wires connected during the measurement.

      ii. If the voltage is less than 10 VDC then proceed to step 6.

4-2 Removal of Fans (Continued)

iii. If the voltage is still greater than 10 VDC then wait the time necessary for it to decrease below 10 VDC before proceeding to step 6.

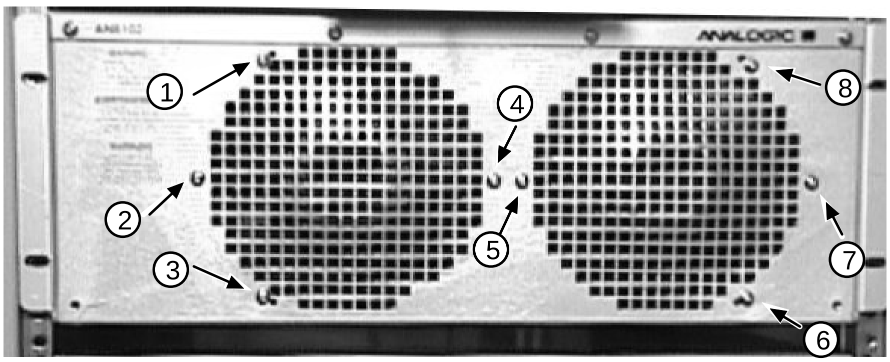
b. If DS8 is not illuminated then use a multimeter to measure the voltage between the exposed portions of the spade connectors of the +HV and HV RTN lines. Do not remove the wires from the spade connectors. Leave the wires connected during the measurement.

i. If the voltage is less than 10 VDC then proceed to step 6.

ii. If the voltage is still greater than 10 VDC then wait the time necessary for it to decrease below 10 VDC before proceeding to step 6.

6. Remove the eight screws holding the fans in place. See Illustration 4-2.

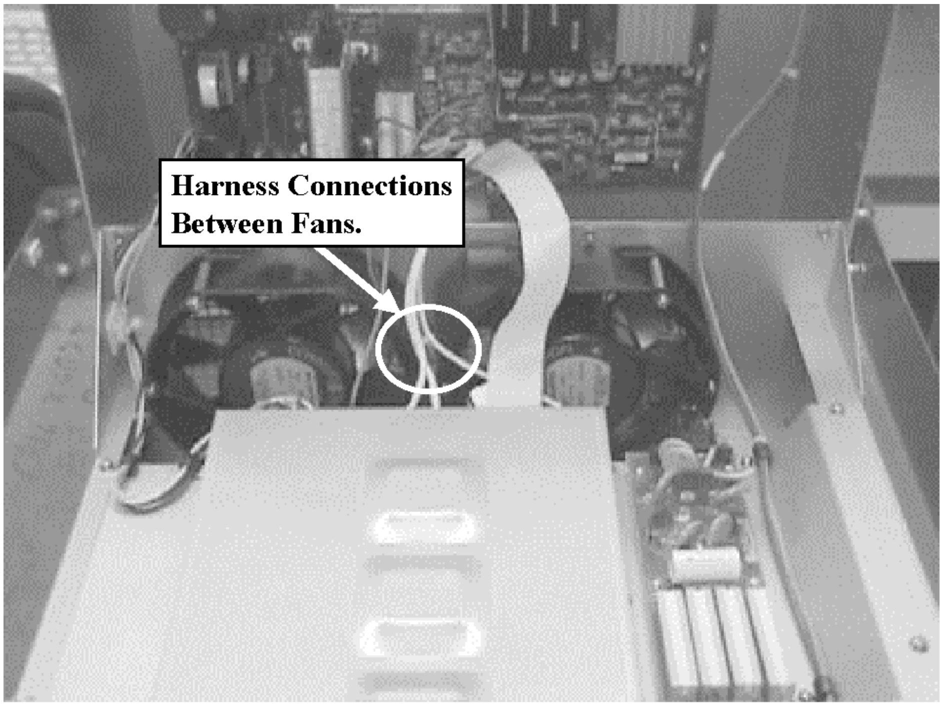
FRONT VIEW



SCREW LOCATIONS FOR FANS
ILLLUSTRATION 4-2

4-2 Removal of Fans (Continued)

7. Lifting one fan at a time, remove fan harness connection from the fans. **Do not pull on wires, grasp by connector.** See Illustration 4-3.



FAN HARNESS LOCATION
ILLUSTRATION 4-3

4-3 Fan Installation

Follow the steps below to install the cooling fans:

1. Install fan harness to connections on fans.

2. Install 2 screws finger-tight in each fan, making sure not to crimp wires between the fan and chassis.

3. Close RF Amplifier top cover assembly and install remaining 6 screws into the fans.

4. Open cover and check again to insure fan harness is not crimped. Close cover and install remaining 16 screws in RF Amplifier top cover assembly.

5. This completes the procedure for installing cooling fans. Proceed to Section **3-2 Amplifier Installation.**

APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR 1.5T SRF CABINETS

A-1 Introduction

WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

The Universal Lift Hoist is assembled from parts stored in the Universal Lift Hoist Kit. Lift attachments specially designed for the 1.5T SRF Cabinet are found in the outer-most compartments of the lift kit. The lift brackets are found in the RF amplifier FRU container.

A-2 Tools Required

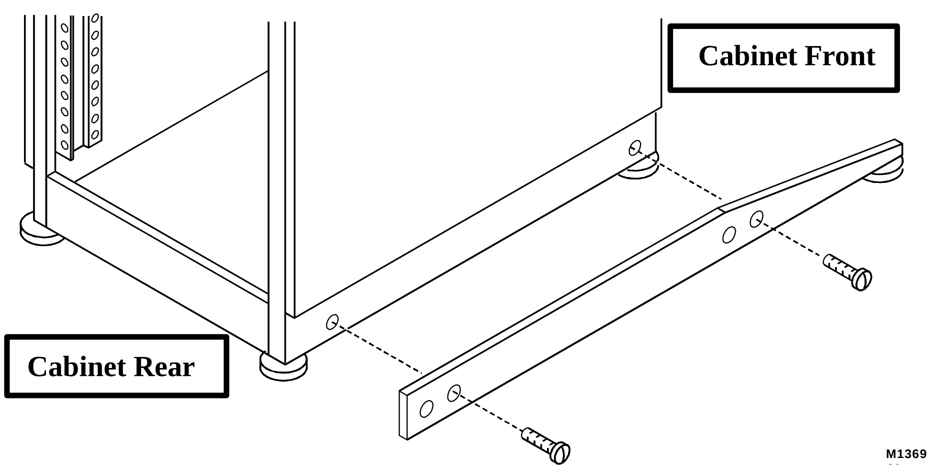
- Universal Lift Hoist Kit with RF/PDU and SRF attachments, 46-317266G5
- Small hand tools including a Phillips-head screwdriver
- Two medium sized crescent wrenches

A-3 Assembly Procedural Steps

DANGER!!

CABINET TIP HAZARD! POSSIBLE PERSONAL INJURY! MAKE SURE THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON EITHER SIDE OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO REMOVE AN RF AMPLIFIER.

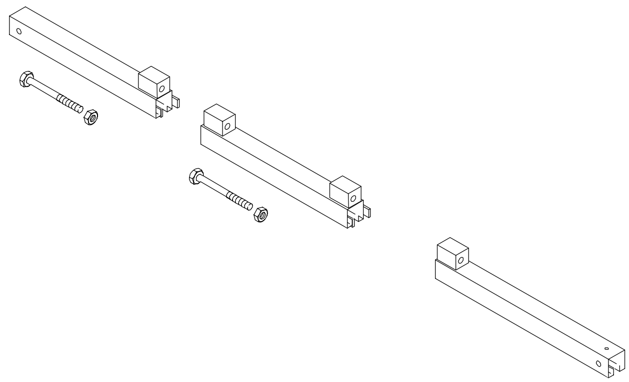
1. Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the SRF Cabinet. See Illustration A-1.



INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET
ILLUSTRATION A-1

A-3 Assembly Procedural Steps (continued)

2. Locate the three (3) large Steel Rail sections and place them on a flat surface for assembly. See Illustration A-2.

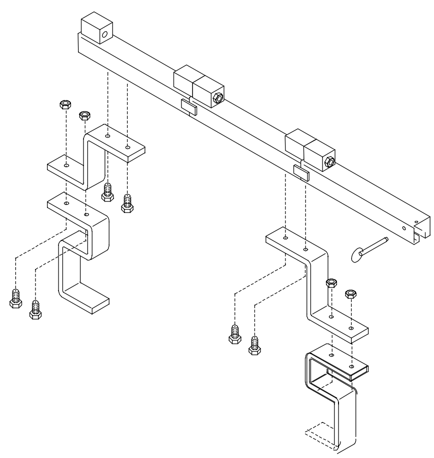


STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS
ILLUSTRATION A-2

3. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75” X 1/2” caps crews and hex nuts.

4. Use the crescent wrenches to tighten the cap screws.

5. Assemble and fasten the elevating brackets to the rail assembly. See Illustration A-3.

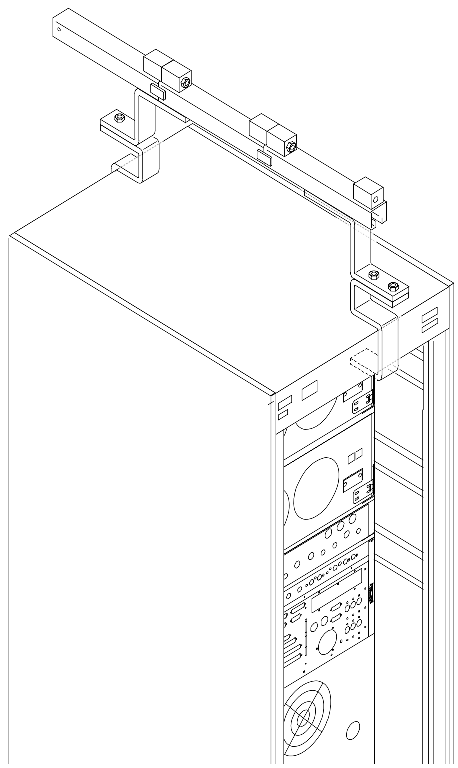


HOIST TOP RAIL PARTS WITH ELEVATING BRACKETS
ILLUSTRATION A-3

6. Remove the front and rear covers of the RF/PDU or SRF Cabinet.

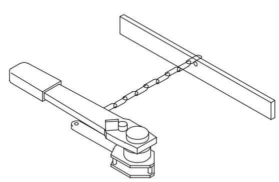
A-3 Assembly Procedural Steps (continued)

7. Place the Steel Rail and elevating bracket assembly at the top of the RF/PDU or SRF Cabinet. Center the Rail Assembly on the top of the RF Amplifier. See Illustration A-4.



HOISTPARTS REAR VIEW ASSEMBLED
ILLUSTRATION A-4

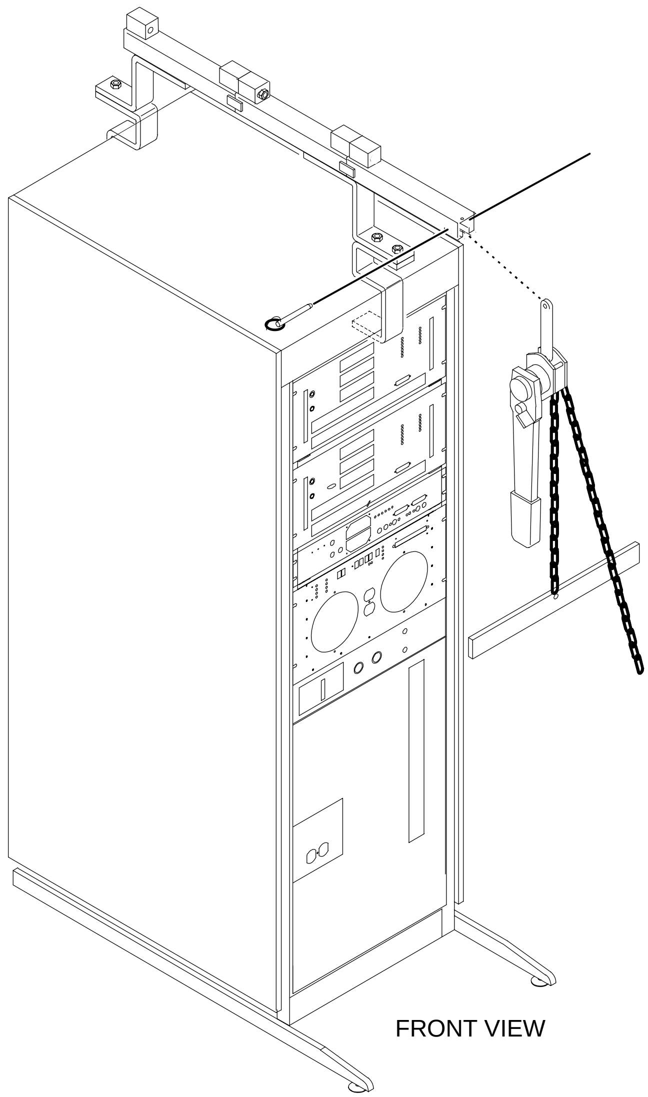
8. Locate the Hoist Assembly in the Lift Kit. See Illustration A-5.



HOIST ASSEMBLY
ILLUSTRATION A-5

A-3 Assembly Procedural Steps (continued)

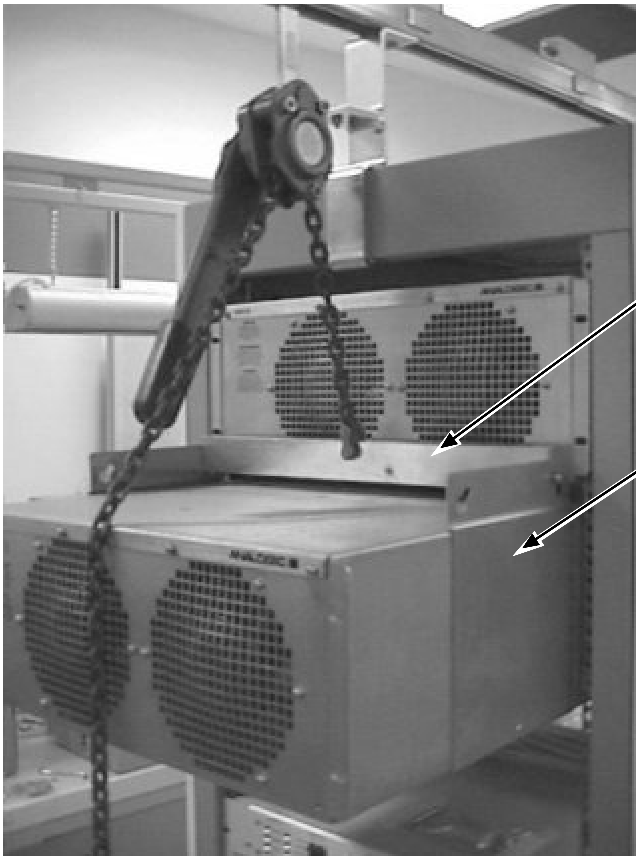
9. Insert the Hoist Assembly into the open end of the front rail. Insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. See Illustration A-6.



PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY
ILLUSTRATION A-6

A-3 Assembly Procedural Steps (continued)

10. Select the proper lift brackets for the Analogic RF Amplifier and attach to side of amplifier, then attach the Hoist Assembly to the lift bracket. See Illustration A-7.



ATTACHING LIFT BRACKET TO HOIST ASSEMBLY
ILLUSTRATION A-7

A-3 Assembly Procedural Steps (continued)

11. The Universal Lift Hoist is now ready for operation. Illustration A-8 shows the Universal Lift Hoist being used to remove a RF Amplifier Module from an RF/PDU Cabinet. Although the location of the RF components in the SRF Cabinet is different, the removal and installation process is the same as for the RF/PDU Cabinet.

USING THE UNIVERSAL LIFT HOIST TO REMOVE THE UPPER RF AMP
ILLUSTRATION A-8

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Dec. 20, 2000	D. Thome’	Initial Release.
1	May 8, 2001	D. Thome’	Added anti-tip brackets. Updated Lockout/Tagout.
2	Aug. 15, 2001	D. Thome’	Updated Lockout/Tagout. Changed Danger messages.
3	Mar. 27, 2003	D. Thome’	Changed fan part number in Table 1-1.
4	10/17/2007	K.Keshena	PSR 13105722, requires the warning statement for the Hoist Kit.